

Web appendix of “Two-way fixed effects estimators with heterogeneous treatment effects”

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Abstract

In this web appendix, we first discuss whether common trends necessarily implies homogeneous treatment effect. Second, we show that the decomposition in Theorem 1 in the paper extends to fuzzy designs, to regressions with covariates, to regressions with a non-binary treatment, and we derive another decomposition under the supplementary assumption that the treatment effect does not change over time. Third, we extend the DID_M estimator to non-binary treatments. Fourth, we discuss inference. Fifth, we review all the papers included in our survey of papers published in the AER between 2010 and 2012 (see Section 6 of the paper). Finally, the last section gathers the proofs of all the additional results in this Web Appendix.

1 Can common trends hold with heterogeneous treatment effects?

Throughout the paper, we assume that groups experience common trends, but that the effect of the treatment may be heterogeneous between groups and / or over time. We now discuss two examples where this may happen. We then argue that the mechanisms behind these examples are fairly general. Thus treatment effects are often likely to be heterogeneous, even when common trends are plausible.

First, assume one wants to learn the effect of the minimum wage on the employment levels of some US counties. For simplicity, let us assume that the minimum wage can only take two values, a low and a high value. Also, let us assume that there are only two periods, the 90s and the 2000s. Between these two periods, the amount of competition from China for the US industry increased substantially. Thus, for the common trends assumption to hold for counties A and B, the effect of that increase in competition should be the same on average in those two

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counties, in the counterfactual state of the world where A and B have a low minimum wage at both dates. For that to be true, the economy of those two counties should be pretty similar. For instance, if A has a very service-oriented economy, while B has a very industry-oriented economy, it is unlikely that their employment levels will react similarly to Chinese competition.

Now, if the economies of A and B are similar, they should also have similar effects of the minimum wage on employment, thus implying that the treatment effect is homogenous between groups. On the other hand, the treatment effect may vary over time. For instance, the drop in the employment levels of A and B due to Chinese competition will probably be higher if their minimum wage is high than if their minimum wage is low. This is equivalent to saying that the effect of the minimum wage on employment diminishes from the first to the second period: due to Chinese competition in the second period, the minimum wage may have a more negative effect on employment then.¹

Second, assume one wants to learn the effect of a job training program implemented in some US counties on participants' wages. Let us suppose that individuals self-select into the training according to a Roy model:

$$D_{i,g,t} = 1\{Y_{i,g,t}(1) - Y_{i,g,t}(0) > c_{g,t}\}, \quad (23)$$

where $c_{g,t}$ represents the cost of the training for individuals in county g and period t . We consider fuzzy designs such as this one in the next section. Here, the common trends condition requires that average wages without the training follow the same evolution in all counties. As above, for this to hold counties used in the analysis should have similar economies, so let us assume that those counties are actually identical copies of each other: at each period, their distribution of wages without and with the training is the same. Therefore, $(g, t) \mapsto E(Y_{g,t}(1) - Y_{g,t}(0))$ is constant. However, $c_{g,t}$ may vary across counties and over time: some counties may subsidize the training more than others, and some counties may change their subsidies over time. Then, $(g, t) \mapsto E(Y_{i,g,t}(1) - Y_{i,g,t}(0) | D_{i,g,t} = 1)$ will not be constant, despite the fact that all counties in the sample have similar economies and experience similar trends on their wages.

Overall, when the treatment is assigned at the group $\times$ period level as in the minimum wage example, the economic restrictions underlying the common trends assumption may also imply homogeneous treatment effect between groups. However, those restrictions usually do not imply that the treatment effect is constant over time. Moreover, when the treatment is assigned at the individual level, as in the job training example, the economic restrictions underlying the common trends assumption neither imply homogeneous treatment effects between groups, nor homogeneous treatment effects over time.

¹ To simplify our discussion, in this example we consider only two counties. But in order to estimate consistently average treatment effects in the presence of county-specific shocks, the number of groups should tend to infinity, as in Section 5 below.

2 Results in fuzzy designs

In this section, the research design may be fuzzy: the treatment may vary within (g, t) cells. For instance, Enikolopov et al. (2011) study the effect of having access to an independent TV channel in Russia, and in each Russian region some people have access to that channel while other people do not.

2.1 Generalizing the decomposition of β_{fe} to fuzzy designs

For any $(g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}$, let

$$\Delta_{g,t}^{TR} = \frac{1}{N_{g,t}D_{g,t}} \sum_{i:D_{i,g,t}=1} [Y_{i,g,t}(1) - Y_{i,g,t}(0)]$$

denote the average treatment effect across the treated units of cell (g, t) . One has

$$\delta^{TR} = E \left(\sum_{g,t} \frac{N_{g,t}D_{g,t}}{N_1} \Delta_{g,t}^{TR} \right),$$

which generalizes (2) to fuzzy designs.² Theorem 1 shows that β_{fe} is also equal to the expectation of a weighted sum of the $\Delta_{g,t}^{TR}$ s. Let

$$w_{g,t}^{TR} = \frac{\varepsilon_{g,t}}{\sum_{g,t} \frac{N_{g,t}D_{g,t}}{N_1} \varepsilon_{g,t}}.$$

Theorem S1 *Suppose that Assumptions 1 and 3-5 hold. Then,*

$$\beta_{fe} = E \left(\sum_{g,t} \frac{N_{g,t}D_{g,t}}{N_1} w_{g,t}^{TR} \Delta_{g,t}^{TR} \right).$$

Theorem S1 shows that in fuzzy designs, β_{fe} is equal to the expectation of a weighted sum of the ATTs in each (g, t) cell. Again, some of the weights may be strictly negative. Note that under Assumption 2, Theorem S1 reduces to Theorem 1 in the paper.

The weights have a simple expression in the following special case.

Assumption S1 (*Heterogenous adoption*) $T = 2$ and for all $g \in \{1, \dots, G\}$, $D_{g,2} > D_{g,1} = 0$.

Assumption S1 is satisfied in applications with two time periods, and where all groups are fully untreated at $t = 1$ and partly treated at $t = 2$. This type of design often arises in practice, for instance when an innovation is heterogeneously adopted by various groups.

²Any equation with a numbering lower than (23) refers to an equation in the paper.

Proposition S1 *If Assumptions 1 and S1 hold and $N_{g,2}/N_{g,1}$ does not vary across g , then*³

$$w_{g,2}^{TR} = \frac{D_{g,2} - D_{.,2}}{\sum_{g=1}^G \frac{N_{g,2}}{N_1} (D_{g,2} - D_{.,2})^2}.$$

Proposition S1 shows that in the heterogeneous adoption design, β_{fe} assigns negative weights to the period-two ATT of groups with a mean treatment lower than the mean treatment in the full population. The reason why negative weights arise is intuitive. With two periods, the FE regression is equivalent to a regression of the first difference of the outcome on the period-two treatment in each group. This regression compares the evolution of the outcome in more- and less-treated groups. Doing so, it subtracts the treatment effect of the less-treated groups, hence the negative weights. Negative weights are a concern if the ATTs of the less- and more-treated groups systematically differ. This could be the case if treatment is determined by a Roy selection model. Then, the groups with the highest proportion of treated units could also be those where the ATT is the highest. On the other hand, if the proportion of treated units is randomly assigned to each group, negative weights are not a concern.⁴

The DID_M estimator can also be generalized to fuzzy designs, see point 2 of Theorem S1 in the Web Appendix of de Chaisemartin and D’Haultfœuille (2018) for further details.

2.2 Application to Enikolopov et al. (2011)

Enikolopov et al. (2011) study the effect of NTV, an independent TV channel introduced in 1996 in Russia, on the share of the electorate voting for opposition parties. NTV’s coverage rate was heterogeneous across subregions: while a large fraction of the population received NTV in urbanized subregions, a smaller fraction received it in more rural subregions. The authors estimate the FE regression: they regress the share of votes for opposition parties in the 1995 and 1999 elections in Russian subregions on subregion fixed effects, an indicator for the 1999 election, and on the share of the population having access to NTV in each subregion at the time of the election. In 1995, the share of the population having access to NTV was equal to 0 in all subregions, while in 1999 it was strictly greater than 0 everywhere. Therefore, the authors’ research design corresponds exactly to the heterogeneous adoption design discussed above. Enikolopov et al. (2011) find that $\hat{\beta}_{fe} = 6.65$ (s.e.= 1.40). According to this regression, increasing the share of the population having access to NTV from 0 to 100% increases the share of votes for the opposition parties by 6.65 percentage points. Because there are only two time periods in the data and the regression is not weighted by subregions’ populations, $\hat{\beta}_{fe} = \hat{\beta}_{fd}$.

We use the `twowayfeweights` Stata package, downloadable with its help file from the SSC repository, to compute the weights attached to $\hat{\beta}_{fe}$. In 1995, all the weights are equal to zero

³Under Assumption S1, $D_{g,1} = 0$, so $w_{g,1}^{TR}$ does not enter in the decomposition in Theorem 1.

⁴Corollaries 1 and 2 extend directly to fuzzy settings.